

Phase Space Reconstruction of Traffic Flow Dynamics

Complex Systems Portfolio Assignment

Daniils Krasovskis 2079606

Tilburg University

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1 Introduction

In this report, we investigate traffic flow dynamics using phase space reconstruction. The goal is to examine whether phase space reconstruction can reveal recurrent and constrained dynamical structure in traffic flow time series arising from processes such as daily activity patterns and congestion effects.

Research Question: To what extent does phase space reconstruction reveal recurrent and constrained dynamical structure in traffic flow time series?

Traffic systems are complex because they are influenced by many interacting processes, including human behaviour, infrastructure constraints, and recurring patterns such as rush hours. These processes create observable patterns in traffic volume over time, but the full state of the system cannot be directly observed. In this study, we analyse traffic volume measured at a single location over time.

Using phase space reconstruction, we can infer hidden system dynamics from a single observed variable (Kantz and Schreiber, 2003). Although only traffic volume is measured directly, the time series still contains information about the evolving state of the traffic system. Delay embedding works because successive observations are temporally dependent rather than independent, meaning that delayed coordinates retain information about previous system states. By combining delayed observations, we can reconstruct the geometry of the system in a higher-dimensional state space.

If the signal is dominated mainly by stochastic fluctuations, the reconstructed phase space is expected to appear diffuse and unstructured. In contrast, recurrent and bounded trajectories suggest temporal dependence and underlying dynamical structure. Using this approach, we examine whether traffic flow exhibits non-trivial structure that cannot be explained by random fluctuations alone.

2 Dataset Description

The dataset consists of traffic volume measurements collected from the Transport Victoria Open Data platform (Department of Transport and Planning, 2026). Traffic volume was recorded at regular 15-minute intervals during March 2026, resulting in a total of 2976 observations.

The dataset contains four variables: timestamp, traffic volume, day of the week, and a weekend indicator. In this analysis, we use the traffic volume variable as a univariate time series for phase space reconstruction. Daily observations were ordered chronologically to form a continuous monthly time series.

The data are uniformly sampled and temporally ordered, making them suitable for non-linear time series analysis and delay embedding. The one-month duration allows recurring temporal patterns, such as daily traffic cycles and transitions between low- and high-traffic regimes, to be observed across multiple repetitions.

3 Methodology

The analysis is based on phase space reconstruction using delay embedding. The main idea behind this approach is that successive observations in a time series are temporally dependent rather than independent. As a result, delayed observations contain partial information about previous system states, allowing the geometry of the underlying dynamics to be reconstructed from a single observed variable.

Given a time series $x(t)$ representing traffic volume, the reconstructed state space is defined as

$$X(t) = (x(t), x(t + \tau), \dots, x(t + (m - 1)\tau)),$$

where τ denotes the time delay and m denotes the embedding dimension.

In this study, the traffic volume variable was analysed as a univariate time series. The observations were temporally ordered and sampled at regular 15-minute intervals. Prior to reconstruction, the traffic volume signal was standardized to allow geometric comparison between the empirical and surrogate reconstructions. Delay embedding was then applied to reconstruct the phase space of the system.

The embedding dimension was set to $m = 3$ primarily for exploratory analysis and visualization of the reconstructed dynamics. The delay parameter was set to $\tau = 10$ observations, corresponding to a temporal shift of approximately 150 minutes.

The choice of τ reflects a balance between redundancy and independence in the delayed coordinates. If τ is too small, neighbouring coordinates remain highly correlated and the reconstructed trajectory does not unfold clearly. If τ is too large, temporal continuity between

coordinates becomes weaker and important dynamical structure may be lost. Preliminary inspection of the time series suggested that delays substantially shorter than 10 observations produced highly redundant embeddings, whereas larger delays weakened continuity in the reconstructed trajectories.

More formal parameter estimation methods, such as average mutual information and false nearest neighbours (Cao, 1997; Wallot and Mønster, 2018), can be used to estimate suitable values for τ and m . However, in this study a heuristic parameter choice was considered sufficient for exploratory nonlinear analysis and visualization.

To provide a simple reference for comparison, a noisy cyclic surrogate signal was generated and reconstructed using the same embedding parameters, following the general logic of surrogate data comparison (Theiler et al., 1992). The surrogate signal was used as a qualitative baseline for comparing the geometric structure of the empirical traffic reconstruction.

A limitation of this methodology is that the reconstruction is based on a single observed variable and heuristic parameter selection. As a result, the reconstructed geometry should be interpreted as an approximation of the underlying system dynamics rather than a complete representation of the full traffic system.

4 Results

This section presents the observed structure of the traffic volume time series and the reconstructed phase spaces obtained through delay embedding.

As shown in Figure 1, the traffic volume signal exhibits clear recurring peaks and troughs corresponding to daily traffic cycles. Higher traffic volumes occur during typical commuting periods, while lower values are observed during nighttime hours. Although the signal displays strong periodic structure, variations in peak intensity and short-term fluctuations indicate that the dynamics are not perfectly regular and may contain stochastic components.

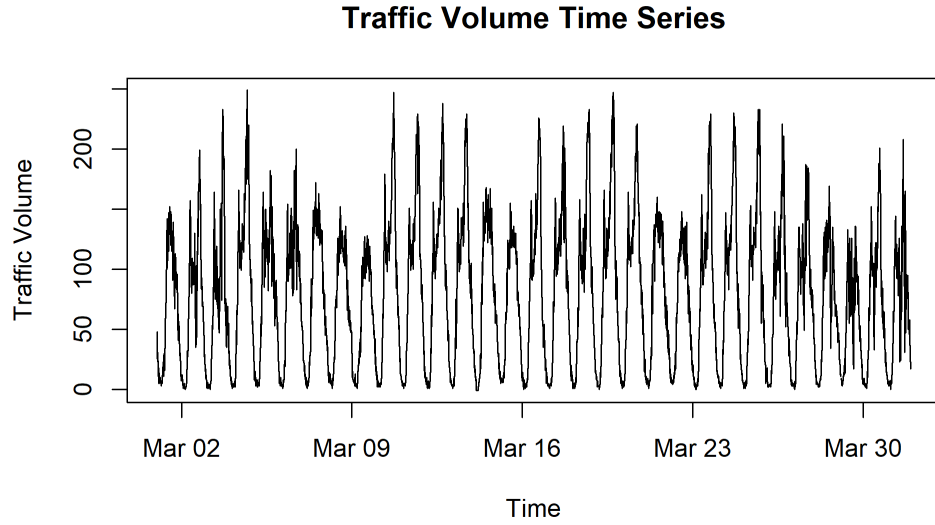


Figure 1: Traffic volume time series for a single location over one month.

Figure 2 shows the autocorrelation function of the traffic volume signal. Strong positive autocorrelation is observed at small lags, indicating substantial temporal dependence between neighbouring observations. The autocorrelation gradually decreases and becomes negative at intermediate lags before increasing again near lag 96, corresponding to approximately 24 hours. This periodic reappearance of positive autocorrelation reflects the presence of recurring daily traffic cycles and supports the use of delayed coordinates for phase space reconstruction.

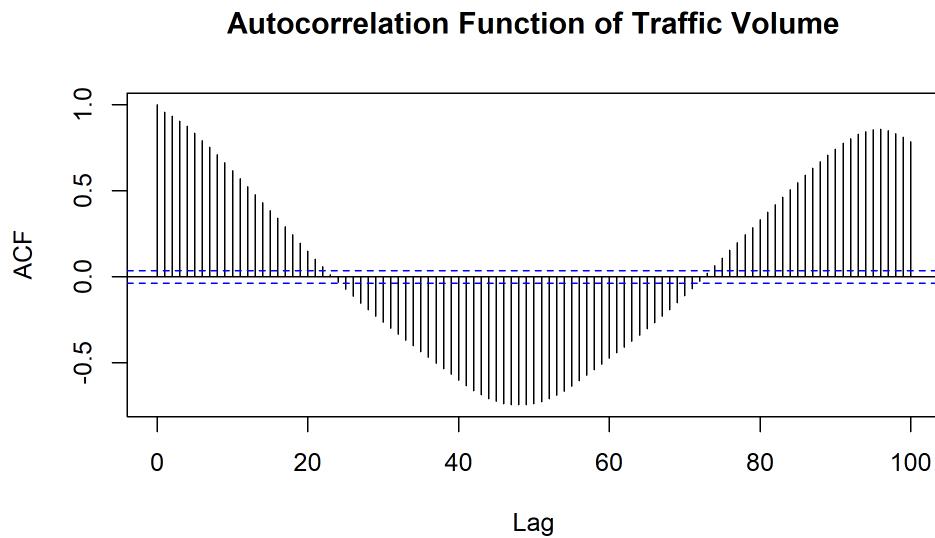


Figure 2: Autocorrelation function of the traffic volume time series.

As illustrated in Figure 3, the two-dimensional reconstructed phase space was created using the standardized traffic volume signal. The reconstruction forms a non-uniform and curved geometric region rather than a completely diffuse point cloud. Dense regions indicate that the system repeatedly visits similar traffic states over time, suggesting recurrence and temporal organization in the signal. The reconstructed trajectories also exhibit continuity between neighbouring states, reflecting gradual transitions between traffic conditions.

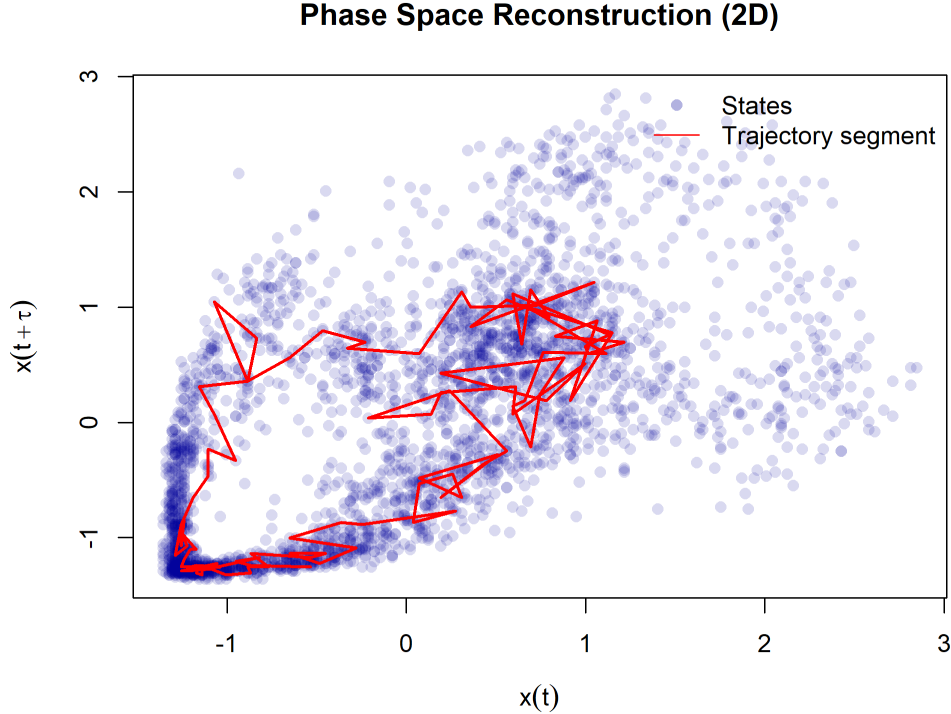


Figure 3: Two-dimensional phase space reconstruction using delay embedding with $\tau = 10$.

To provide a reference for comparison, a noisy cyclic surrogate signal was generated, standardized, and reconstructed using the same embedding parameters. As shown in Figure 4, the surrogate reconstruction exhibits broader and more diffuse geometric structure compared to the empirical traffic signal. Although both signals contain cyclic organization, the empirical traffic reconstruction appears more coherent and recurrent, suggesting that the observed structure is not explained solely by periodicity and random noise.

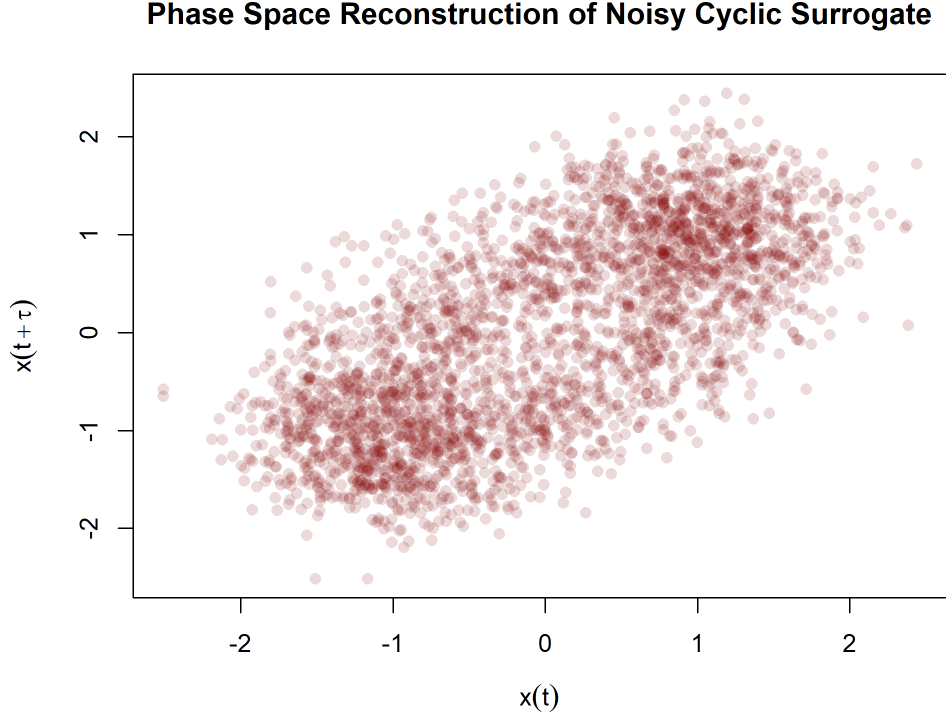


Figure 4: Phase space reconstruction of a noisy cyclic surrogate signal.

5 Discussion

The analyses suggest that the traffic volume signal contains substantial temporal structure rather than behaving as a sequence of independent random observations. The autocorrelation analysis revealed strong temporal dependence at small lags and recurring positive correlations at approximately 24-hour intervals, indicating recurring daily traffic dynamics.

The reconstructed phase space provides a geometric representation of these temporal relationships. Instead of forming a diffuse and structureless point cloud, the empirical reconstruction exhibits recurrent trajectories and locally dense regions corresponding to repeatedly visited traffic states. Comparison with the noisy cyclic surrogate signal further suggests that the observed geometry of the traffic signal is not explained solely by periodicity and random noise.

At the same time, the results should not be interpreted as evidence of a fully deterministic system. The reconstruction depends strongly on the selected embedding parameters and is based on a single observed variable from one traffic location.

In this study, the delay parameter $\tau = 10$ was chosen because shorter delays produced highly correlated embeddings, while substantially larger delays weakened temporal continuity

in the reconstructed trajectories. The embedding dimension $m = 3$ was selected primarily for exploratory visualization of the reconstructed dynamics.

More rigorous parameter estimation methods, such as average mutual information and false nearest neighbours (Cao, 1997; Wallot and Mønster, 2018), could provide more systematic estimates of suitable embedding parameters. In addition, the analysis was limited to a single traffic location and one observed variable, meaning that many spatial and external influences on traffic dynamics were not directly captured.

Although the comparison with the surrogate signal suggests additional temporal organization within the empirical traffic signal, the present analysis remains primarily exploratory and qualitative. Further nonlinear analyses, such as recurrence quantification analysis or Lyapunov exponent estimation, could provide more quantitative characterization of the reconstructed dynamics.

Overall, the study demonstrates how phase space reconstruction can provide insight into temporal organization and recurrence in real-world systems using only a single observed variable.

6 Conclusion

This study applied phase space reconstruction to traffic volume data in order to investigate whether the observed dynamics exhibit temporal structure beyond random fluctuations. The results revealed strong temporal dependence, recurring daily cycles, and coherent reconstructed geometric structure within the phase space.

Comparison with a noisy cyclic surrogate signal further suggested that the observed structure of the traffic signal is not explained solely by periodicity and random noise. Instead, the reconstructed trajectories exhibited stronger recurrence and temporal organization than the surrogate reconstruction.

Overall, the results suggest that phase space reconstruction is capable of revealing recurrent and temporally constrained structure in traffic flow time series, even when only a single observed variable is available. The method provided insight into temporal dependence, state recurrence, and geometric organization within the traffic signal beyond what would be expected from random fluctuations alone.

AI Usage Statement

AI tools were used for language editing, structuring ideas, and debugging code. The analysis, methodological choices, interpretation of results, and final written content were reviewed and developed by the author.

References

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